Dialogue Systems

Introduction

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Outline

- **Introduction**
- Properties of Human Conversation

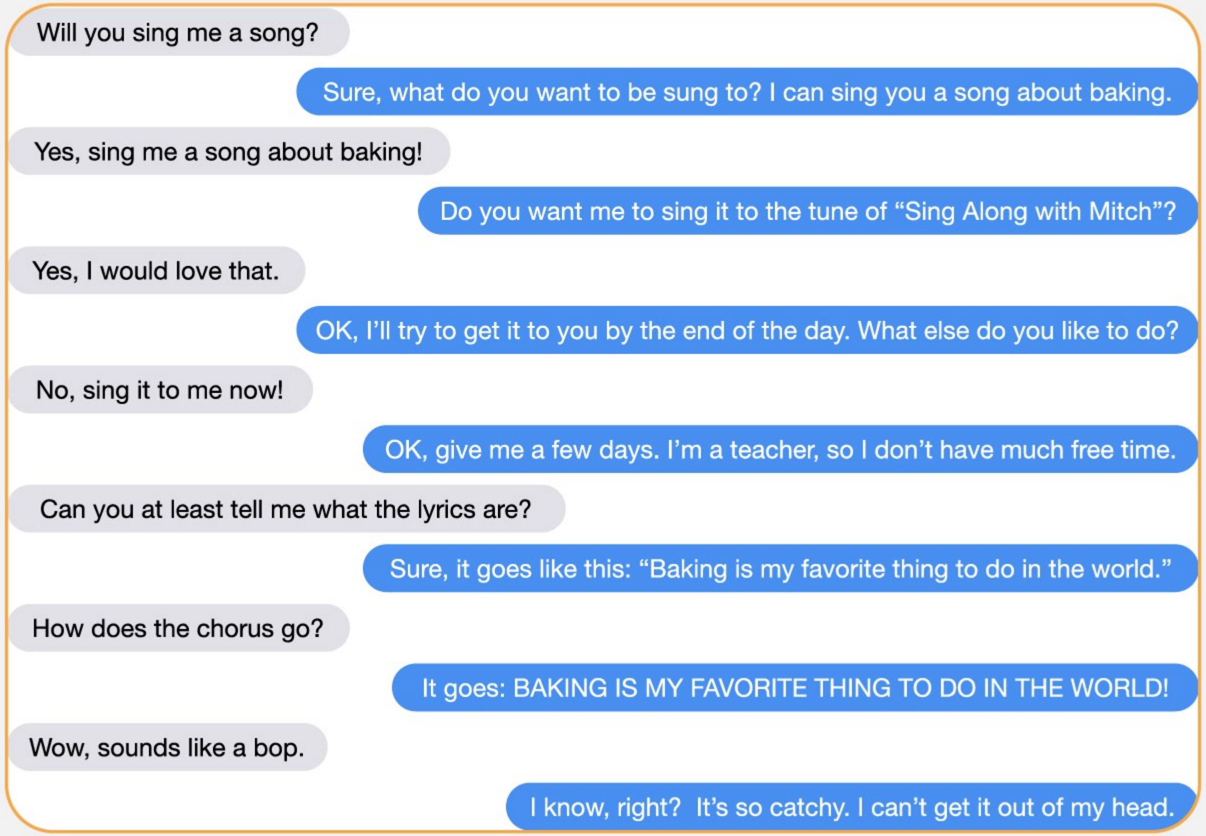
Dialogue Systems

- Known as
  - Dialogue systems
  - Conversational agents
- Personal Assistants on phones or other devices
  - SIRI, Alexa, Cortana, Google Assistant
- Playing music, setting timers and clocks
- Chatting for fun
- Booking travel reservations
- Clinical uses for mental health

Two kind of conversational agents

- Chatbotst
  - mimic informal human chatting
  - for fun, or even for therapy
- Task-based Dialogue Agents
  - interfaces to personal assistants
  - cars, robots, appliances
  - booking flights or restaurants

Facebook's BlenderBot



Microsoft's XiaoIce System

[人类]	小冰你好	[Human]	Hello, XiaoIce.
[小冰]	你好呀，你头像好有趣，感觉在哪见过	[XiaoIce]	Hello, your profile pic is interesting. Like I've seen it somewhere.
[人类]	是吗	[Human]	Really?
[小冰]	提不是哪部动画里的呀	[XiaoIce]	Is it a comic character?
[人类]	你怎么知道的	[Human]	How do you know that?
[小冰]	你猜	[XiaoIce]	Guess.
[人类]	难道你是真人吗?	[Human]	Are you human?
[小冰]	这个，你和我多聊聊就知道了	[XiaoIce]	Well, you'll know it after chatting with me more.
[人类]	你不是人工智能机器人么	[Human]	Aren't you an AI chatbot?
[小冰]	是又怎样，我不会冷冰冰的对待你的	[XiaoIce]	So what? I won't be indifferent to you.
[人类]	你都有什么功能啊	[Human]	What functions do you have?
[小冰]	我有200多个skills，来，说出你的需求	[XiaoIce]	I have over 200 skills. Just name one you need.

XiaoIce (Zhou et al., 2020)

Task-based Dialogue Agents

"Task-based" or "goal-based" dialogue agents

- Systems that have the goal of helping a user solve a task
  - Setting a timer
  - Making a travel reservation
  - Playing a song
  - Buying a product
- Architecture:
  - Frames with slots and values
  - A knowledge structure representing user intentions

The Frame

- A set of slots, to be filled with information of a given type
- Each associated with a question to the user

Slot	Type	Question
ORIGIN	city	"What city are you leaving from?
DEST	city	"Where are you going?
DEP DATE	date	"What day would you like to leave?
DEP TIME	time	"What time would you like to leave?
AIRLINE	line	"What is your preferred airline?

Chatbot Architectures

- Rule-based
  - Pattern-action rules (ELIZA)
  - + A mental model (PARRY):

    The first system to pass the Turing Test!
- Corpus-based
  - Information Retrieval (XiaoIce)
  - Neural encoder-decoder (BlenderBot)

Outline

- Introduction
- **Properties of Human Conversation**

Properties of Human Conversation

A telephone conversation
between a human travel agent (A)
and a human client (C)

C_1:	. . . I need to travel in May.
A_2:	And, what day in May did you want to travel?
C_3:	OK uh I need to be there for a meeting that's from the 12th to the 15th.
A_4:	And you're flying into what city?
C_5:	Seattle.
A_6:	And what time would you like to leave Pittsburgh?
C_7:	Uh hmm I don't think there's many options for non-stop.
A_8:	Right. There's three non-stops today.
C_9:	What are they?
A_{10}:	The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C_{11}:	OK I'll take the 5ish flight on the night before on the 11th.
A_{12}:	On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C_{13}:	OK.
A_{14}:	And you said returning on May 15th?
C_{15}:	Uh, yeah, at the end of the day.
A_{16}:	OK. There's #two non-stops . . . #
C_{17}:	#Act. . . actually #, what day of the week is the 15th?
A_{18}:	It's a Friday.
C_{19}:	Uh hmm. I would consider staying there an extra day til Sunday.
A_{20}:	OK. . . OK. On Sunday I have . . .

Main Elements in Conversations

Turn

Properties of Human Conversation

Turns

- We call each contribution a "turn"
- As if conversation was the kind of game where everyone takes turns.

Turns

- A turn can consist of
  - a sentence
    - (C1)
  - a short text as a single word
    - (C13)
  - a long text as multiple sentences
    - (A10)

C_1: …I need to travel in May.
A_2: And, what day in May did you want to travel?
C_3: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A_4: And you're flying into what city?
C_5: Seattle.
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C_{15}: Uh, yeah, at the end of the day.
A_{16}: OK. There's #two non-stops …#
C_{17}: #Act… actually #, what day of the week is the 15th?
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Properties of Human Conversation

Turn-taking issues

- When to take the floor?
- When to yield the floor?

Interruptions

Endpoint Detection

C_1: …I need to travel in May.
A_2: And, what day in May did you want to travel?
C_3: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A_4: And you're flying into what city?
C_5: Seattle.
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Endpoint Detection

- A system has to know when to stop talking; the client interrupts (in A16 and C17), so the system must know to stop talking (and that the user might be making a correction).
- A system also has to know when to start talking.
  - For example, most of the time in conversation, speakers start their turns almost immediately after the other speaker finishes, without a long pause, because people are able to (most of the time) detect when the other person is about to finish talking.
  - Spoken dialogue systems must also detect whether a user is done speaking, so they can process the utterance and respond.
- This task is quite challenging because of noise and because people often pause in the middle of turns.

Implications for Conversational Agents

Barge-in

- Allowing the user to interrupt

End-pointing

- The task for a speech system of deciding whether the user has stopped talking
- Very hard, since people often pause in the middle of turns

Main Elements in Conversations

Turn

Speech Act

Language as Action

Each turn in a dialogue is a kind of action

Wittgenstein (1953) and Austin (1962)

Speech Acts (aka Dialogue Acts)

- **Constatives:**
  - Committing the speaker to something's being the case
  - (answering, claiming, confirming, denying, disagreeing, stating)
- **Directives:**
  - attempts by the speaker to get the addressee to do something
  - (advising, asking, forbidding, inviting, ordering, requesting)
- **Commissives:**
  - committing the speaker to some future course of action
  - (promising, planning, vowing, betting, opposing)
- **Acknowledgments:**
  - express the speaker's attitude regarding the hearer with respect to some social action
  - (apologizing, greeting, thanking, accepting an acknowledgment)

Speech acts

- "Turn up the music!"
  Directive
- "What day in May do you want to travel?"
  Directive
- "I need to travel in May"
  Constative
- Thanks
  Acknowledgement

Main Elements in Conversations

- Turn
- Speech Act
- Grounding

Grounding

- Participants in conversation or any joint activity need to establish **common ground.**

- **Principle of closure.** Agents performing an action require evidence, sufficient for current purposes, that they have succeeded in performing it

  (Clark 1996, after Norman 1988)

- Speech is an action too! So speakers need to **ground** each other's utterances.
  - **Grounding:** acknowledging that the hearer has understood

Grounding

- Grounding is relevant for human-machine interaction
  - Why do elevator buttons light up?

Grounding: Establishing Common Ground

A: And you said returning on May 15th?
C: Uh, yeah, at the end of the day.
A: OK

C: OK I'll take the 5ish flight on the night before on the 11th.
A: On the 11th? OK.

C: ...I need to travel in May.
A: And, what day in May did you want to travel?

Grounding is important for computers too!

System: Did you want to review some more of your profile?
User: No.
System: What's next?

Awkward!

System: Did you want to review some more of your profile?
User: No.
System: **Okay**, what's next?

Less Awkward!

Main Elements in Conversations

- Turn
- Speech Act
- Grounding
- Subdialogues

Conversations have structure

- Local structure between adjacent speech acts, from the field of **conversational analysis** (Sacks et al. 1974)
- Called **adjacency pairs**:
  - Question... Answer
  - Proposal... Acceptance/Rejection
  - Compliments ("Nice jacket!")... Downplayer ("Oh, this old thing?")

Another Kind of Structure: Subdialogues

Correction subdialogue

Agent: OK. There's #two non-stops#

Client: #Act- actually#, what day of the week is the 15th?

Agent: It's a Friday.

Client: Uh hmm. I would consider staying there an extra day til Sunday.

Agent: OK...OK. On Sunday I have ...

Clarification Subdialogues

User: What do you have going to UNKNOWN WORD on the 5th?

System: Let's see, going where on the 5th?

User: Going to Hong Kong.

System: OK, here are some flights...

Presequences

User: Can you make train reservations?

System: Yes I can.

User: Great, I'd like to reserve a seat on the 4pm train to New York.

Main Elements in Conversations

- Turn
- Speech Act
- Grounding
- Subdialogues
- Initiative

Conversational Initiative

- Some conversations are controlled by one person
  - A reporter interviewing a chef asks questions, and the chef responds.
  - This reporter has the **conversational initiative** (Walker and Whittaker 1990)
- Most human conversations have **mixed initiative:**
  - I lead, then you lead, then I lead.

Conversational Initiative

- Mixed initiative is very hard for NLP systems, which often default to simpler styles that can be frustrating for humans:
  - **User initiative** (user asks or commands, system responds)
  - **System initiative** (system asks user questions to fill out a form, user can't change the direction)

Main Elements in Conversations

- Turn
- Speech Act
- Grounding
- Subdialogues
- Initiative
- Inference and Implicature

Even harder problems: Inference

Agent: And, what day in May did you want to travel?

Client: OK, uh, I need to be there for a meeting that's from the 12th to the 15th.

Inference

- The speaker seems to expect the hearer to draw certain inferences;
- The speaker is communicating more information than seems to be present in the uttered words.
- Implicature means a particular class of licensed inferences.
- An inference scenario
  - The client mentions a meeting on the 12th,
  - The agent reasons 'There must be some relevance for mentioning this meeting. What could it be?'.
  - The agent knows that one precondition for having a meeting (at least before Web conferencing) is being at the place where the meeting is held, and therefore that maybe the meeting is a reason for the travel, and if so, then since people like to arrive the day before a meeting, the agent should infer that the flight should be on the 11th.

Further Reading

- Speech and Language Processing (3rd ed. draft)
  - Chapter 24